



Systems, DevOps & Platform Engineering Learning Roadmap

Bilingual, hands-on learning journal focused on systems, security, automation, cloud infrastructure, reliability, and platform engineering.



Actively Maintained

Level 1 — Foundations of Systems & Security

Turkish & English

Module 01 — Advanced Networking for Platforms



About This Repository

English

This repository is an ongoing learning journal and practical roadmap created to build a strong technical foundation in system administration, networking, cybersecurity, DevSecOps, cloud computing, Site Reliability Engineering, and platform engineering.

The roadmap combines technical documentation, security-focused analysis, practical validation exercises, and progressively advanced portfolio projects. Content is prepared in both Turkish and English and is updated throughout the learning process.

Türkçe

Bu depo; sistem yönetimi, ağ teknolojileri, siber güvenlik, DevSecOps, bulut bilişim, Site Reliability Engineering ve platform mühendisliği alanlarında güçlü bir teknik temel oluşturmak amacıyla hazırladığım, devam eden bir öğrenme günlüğü ve uygulama yol haritasıdır.

Yol haritası; teknik dokümantasyonlar, güvenlik odaklı değerlendirmeler, uygulamalı doğrulama çalışmaları ve seviyeler ilerledikçe geliştirilecek kapsamlı portföy projelerinden oluşmaktadır. İçerikler Türkçe ve İngilizce olarak hazırlanmakta ve öğrenme süreci boyunca düzenli olarak güncellenmektedir.



Current Progress

Field	Current Status
Current Level	Level 1 — Foundations of Systems & Security
Current Module	Module 01 — Advanced Networking for Platforms
Repository Status	Actively Maintained

Documentation Languages	Turkish and English
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Completion indicators represent the current learning progress. Direct links are added as the corresponding documents are committed to the repository.

Roadmap Overview

Level	Main Focus	Modules	Status
1	Systems, Networking & Security Foundations	01–03	 In Progress
2	Application Infrastructure, Containers & IaC	04–07	 Planned
3	Software Engineering, Automation, CI/CD & Observability	08–11	 Planned
4	Cloud, Reliability, Performance & Intelligent Operations	12–15	 Planned
5	AI Systems Security, MLSecOps & Governance	16–25	 Planned

Detailed Learning Roadmap

Level 1 — Foundations of Systems & Security · In Progress

Module 01 — Advanced Networking for Platforms · In Progress

-  Physical & Data Link Layers — OSI ve TCP/IP modelleri, encapsulation, fiziksel sinyaller, Ethernet frame yapısı, MAC addressing, switching, MTU, interface counters ve Layer 1/2 troubleshooting.
-  ARP & Local Network Dynamics — ARP çalışma mantığı, neighbor table, ARP cache durumları, gratuitous ARP, proxy ARP, ARP spoofing riskleri ve Dynamic ARP Inspection.
-  VLAN & VXLAN — Layer 2 segmentasyonu, access ve trunk portlar, IEEE 802.1Q tagging, VLAN hopping riskleri, VXLAN overlay mimarisi, VTEP ve VNI kavramları.
-  Network Routing & Subnetting — IPv4 adresleme, CIDR, subnetting, routing table, default gateway, static ve dynamic routing temelleri, OSPF/BGP genel mantığı ve IP fragmentation.
-  Transport Layer & Traffic Control — TCP ve UDP, portlar ve socket'ler, TCP three-way handshake, sequence ve acknowledgment numbers, flow control, congestion control, retransmission ve connection termination.
-  Application Layer Services — DNS kayıt türleri, DNS resolution, DNS spoofing riskleri, DHCP DORA süreci, DHCP Snooping, HTTP/HTTPS ve TLS handshake.
-  Load Balancing & Proxy Concepts — Forward proxy ve reverse proxy, Layer 4 ve Layer 7 load balancing, health checks, session persistence, TLS termination ve TLS passthrough.

Module 02 — Enterprise Linux System Administration · In Progress

-  Linux Architecture, Boot Process & FHS — Kernel, initramfs, GRUB, systemd, boot süreci ve Linux izin hiyerarşisi; /etc, /var, /home, /opt, /tmp, /proc, /sys ve /dev.
-  Shell, CLI & Stream Processing — Standard input/output, pipes, redirection, environment variables ve grep, awk, sed, cut, sort, uniq, find, xargs gibi araçlar.
-  User & Permission Management — User ve group yönetimi, UID/GID, chmod, chown, umask, POSIX ACLs, sudoers ve least-privilege yaklaşımı.
-  Process & Resource Management — Process lifecycle, signals, foreground/background jobs, ps, top, htop, kill, file descriptors, memory ve CPU resource takibi.
-  systemd, Services & Scheduled Tasks — Unit files, service dependencies, systemctl, journalctl, restart policies, systemd timers ve cron.
-  Storage, Filesystems & LVM — Partitioning, block devices, ext4, XFS, mount işlemleri, /etc/fstab, inode yapısı, swap, LVM, RAID ve disk kapasite yönetimi.

-  **Linux Networking & Troubleshooting** — Interface yönetimi, routing, DNS çözümüleme, /etc/hosts, systemd-resolved, ip, ss, tcpdump, dig, curl, nc ve network namespaces.
-  **Package & Repository Management** — apt, dpkg, dnf, rpm, repository yapıları, package signatures, dependency management ve güvenli güncelleme süreçleri.
-  **Logging, Time & Operational Maintenance** — /var/log, systemd journal, rsyslog, log rotation, NTP, chrony, backup, restore validation ve bakım planları.

Module 03 — Security Engineering & Incident Management · **Planned**

-  Security Foundations & Risk Management
-  Threat Modeling & Attack Lifecycle
-  Network and Availability Threats
-  Malware & Endpoint Defense
-  Firewalling & Network Defense
-  Cryptography, PKI & SSH Security
-  Linux Hardening & Vulnerability Management
-  System Auditing, Logging & SIEM
-  Log-Based Abuse Mitigation
-  Incident Response Fundamentals

Level 2 — Application Infrastructure, Containers & Infrastructure as Code · **Planned**

- **Module 04 — Web, API & Database Foundations:** Web architecture, HTTP/HTTPS, TLS, cookies, REST APIs, OAuth 2.0, database fundamentals, transactions, indexes, backup, and Redis.
- **Module 05 — Container Technologies — Docker Deep Dive:** Container internals, Dockerfile practices, storage, networking, compose, registries, image lifecycle, and container security hardening.
- **Module 06 — Kubernetes Orchestration & Platform Fundamentals:** Kubernetes architecture, core objects, scheduling, services, storage, RBAC, traffic management, scaling, packaging, troubleshooting, and GitOps.
- **Module 07 — Infrastructure as Code & Configuration Management:** Terraform, modules, remote state security, validation, Ansible playbooks, Vault, Git integration, and Terraform-Ansible operational separation.

Level 3 — Software Engineering, Automation, CI/CD & Observability · **Planned**

- **Module 08 — CI/CD Pipelines & GitOps:** Pipelines, build-test workflows, artifact delivery, DevSecOps gates, OIDC-based cloud auth, supply-chain security, and rollout strategies.
- **Module 09 — Systems Automation & Scripting:** Bash and Python for operations, filesystems, processes, APIs, network checks, secure automation, testing, and packaging.
- **Module 10 — Software Engineering for Infrastructure & Platform Teams:** Clean code, SOLID, patterns, testing, refactoring, architecture, distributed systems, API design, and technical documentation.
- **Module 11 — Observability, Logging & Monitoring:** Metrics, logs, traces, Prometheus, Grafana, Loki, OpenTelemetry, alerting, SLO monitoring, and troubleshooting.

Level 4 — Cloud Computing, Reliability & Intelligent Operations · **Planned**

- **Module 12 — Cloud Computing — AWS Focus:** AWS global infrastructure, IAM, VPC, EC2, ALB/NLB, S3/EBS/EFS, RDS, Lambda, security services, monitoring, DR, governance, and IaC integration.
- **Module 13 — Site Reliability Engineering & High Availability:** SLI/SLO/SLA, toil reduction, incident management, resilience patterns, HA design, disaster recovery, chaos engineering, and operational readiness.
- **Module 14 — Performance Engineering & Capacity Analysis:** CPU, memory, I/O, network performance, profiling, benchmarking, capacity planning, backpressure, and cost-performance trade-offs.
- **Module 15 — AI-Assisted Engineering & Operational Intelligence:** Prompt design, AI-assisted code review, infra review, incident support, verification, hallucination risks, and responsible AI use.

Level 5 — AI Systems Security, MLSecOps & Governance · **Planned**

This level covers advanced AI security engineering topics including:

- **Module 16:** AI, ML & LLM Foundations for Security Engineers
- **Module 17:** AI Threat Modeling & Security Architecture
- **Module 18:** LLM, RAG & Agentic Application Security
- **Module 19:** Adversarial Machine Learning & Model Security
- **Module 20:** AI Data Security, Privacy & Knowledge Protection
- **Module 21:** Secure AI Infrastructure & Model Serving
- **Module 22:** MLSecOps, AI Supply Chain & Secure AI SDLC
- **Module 23:** AI Security Testing, Evaluation & Red Teaming
- **Module 24:** AI Security Monitoring, Threat Detection & Incident Response
- **Module 25:** AI Governance, Assurance, Standards & Compliance

This final level is designed as an advanced specialization in secure AI systems, safe model lifecycles, LLM/RAG/agent security, monitoring, governance, and compliance.

Milestone Portfolio Projects

The following progressively advanced portfolio projects are planned to validate knowledge acquired throughout the roadmap.

Project	Planned Milestone	Focus
Project 1 — Hardened & Audited Linux Server Infrastructure	After Level 1	Host hardening, firewalling, SSH security, Fail2Ban, auditd, logging, backup, Bash validation
Project 2 — Automated Provisioning of a Production-Oriented Kubernetes Environment	After Level 2	Terraform, Ansible, containerd, kubeadm, AWS VPC/EC2/S3, RBAC, NetworkPolicy, rebuild validation
Project 3 — Resilient Cloud-Native GitOps & Observability Platform	After Levels 3 and 4	GitHub Actions, EKS, Argo CD, Prometheus, Grafana, Loki, OpenTelemetry, SLOs, DR, resilience testing



Documentation Approach

- Turkish and English technical documentation
- Architecture and data-flow diagrams
- Security risks and defensive controls
- Hardening recommendations and their limitations
- Command and configuration examples
- Troubleshooting methodology
- Practical validation steps
- Production and laboratory environment distinctions
- Official references and standards

The goal is not to create a separate large laboratory project for every topic. Small exercises are included when they improve understanding. Broader implementations are consolidated into milestone portfolio projects so related technologies can be demonstrated together within realistic system boundaries.

Project Status Notice

This repository represents an ongoing learning process. Completed entries describe topics that have been studied and reviewed. Planned modules and project descriptions define the intended scope of future work.

Technologies, architectures, implementation details and project boundaries may be revised as the systems are built, tested, measured and documented. A planned feature should not be interpreted as an already implemented or validated capability.

Built as a long-term technical learning journal. Scope and implementation details evolve through study, testing, measurement, and documentation.